Behavioral Segmentation for Targeted Marketing

Project Id:10

1. **Student name**: Abhinesh Gangwar (Leader)
   - University Roll number: 2028614
   - Section: G
2. **Student name**: Deepanshi Agarwal
   - University Roll number: 2028647
   - Section: G
3. **Student name**: Avika Khanna
   - University Roll number: 2028682
   - Section: G

1. **Problem Description** 3
2. **Architecture** 4
3. **Models model** 5
4. **Evaluation metrics used** 6
5. **Preliminary Results and Visualizations** 7
6. **References** 8

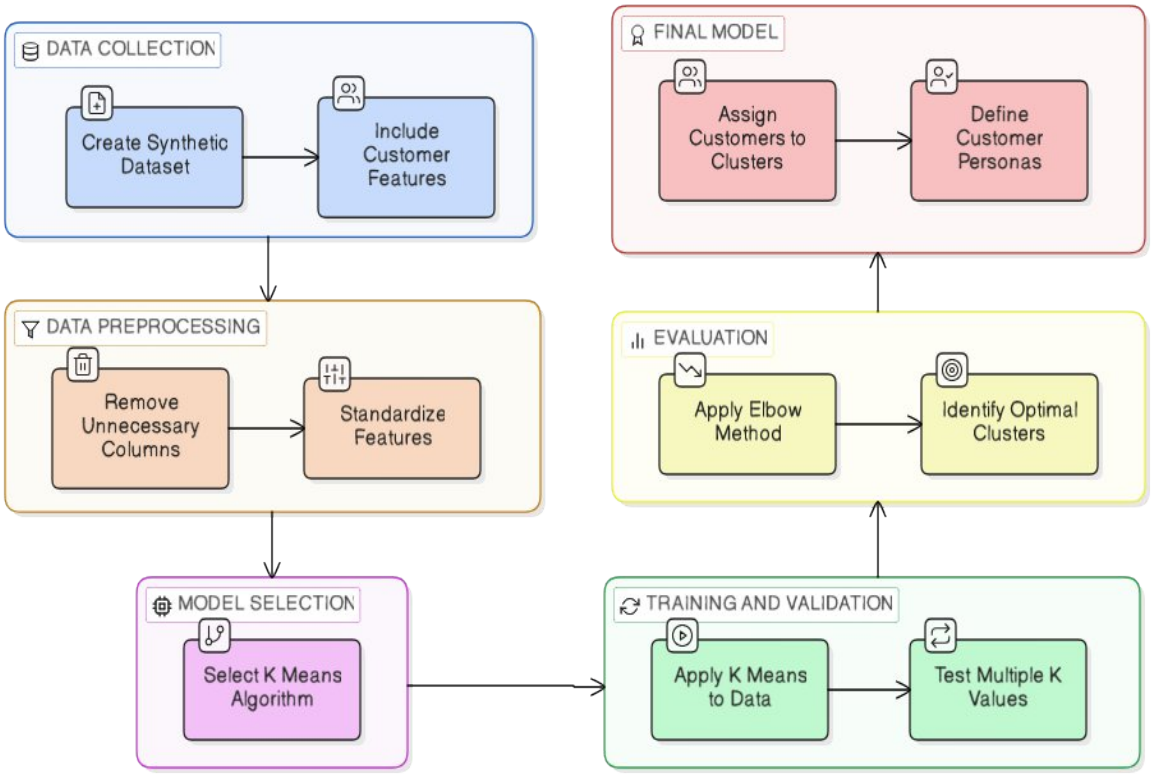
1. Problem Description

Behavioral Segmentation for Targeted Marketing is a machine learning-based system that sorts customers into groups based on how they buy things. The system looks for patterns in customer data like annual income, annual spending, and how often they buy things. Businesses often have trouble figuring out what their customers want and how to best target their ads. Businesses can use clustering algorithms like K-Means to group customers into useful groups with the help of this project. This helps businesses make marketing plans that are tailored to each customer, which makes customers happier and boosts overall sales. The system automates data preprocessing, model training, and visualization to give you accurate and quick segmentation results.

.

2. Architecture

Fig. Architecture diagram



3. Models Used(Applied/ Planned) Give a brief introduction of the models used:

K-Means Clustering is an unsupervised machine learning algorithm used to partition data into distinct groups based on similarity among data points. It is widely used for pattern recognition and customer segmentation because it does not require labelled data. In this project, K-Means is applied to segment customers according to key behavioral features such as annual income, annual spending, and purchase frequency. These features help in understanding customer value and buying behavior more effectively.

The algorithm operates by initially selecting random cluster centroids and then assigning each data point to the nearest centroid based on distance, typically using Euclidean distance. After assignment, the centroids are recalculated as the mean of all points in each cluster. This process is repeated iteratively until the centroids stabilize and an optimal grouping is achieved.

To determine the appropriate number of clusters, the Elbow Method is used, which analyzes inertia values and identifies the point where adding more clusters results in diminishing returns. Additionally, StandardScaler is applied to normalize the dataset so that all features are on the same scale. This prevents any single feature from dominating the clustering process, ensuring more balanced, accurate, and meaningful customer segmentation for targeted marketing strategies.

4. Evaluation metrics used

The Silhouette Score is used to judge how well the clustering model works. The Silhouette Score shows how similar a data point is to its own cluster compared to other clusters. This shows how well the clusters are separated and how well each cluster works together. A higher silhouette score means that the clusters are better defined and more distinct. A lower score means that the clusters are overlapping or not well separated. The Elbow Method is also used to find the best number of clusters. It works by graphing the inertia (the sum of squares within each cluster) against different values of K and finding the point where the drop in inertia starts to slow down a lot. This elbow point helps you choose the right number of clusters for your dataset.

5. Preliminary Results and Visualizations

The preliminary results include various visualizations used to understand customer behavior. A boxplot is used to detect outliers in the dataset. A heatmap is generated to show correlation between features such as income and spending. The Elbow graph helps determine the optimal number of clusters. Scatter plots are used to visualize customer segments and cluster centroids. These visualizations help in interpreting cluster patterns and understanding customer groups**.**

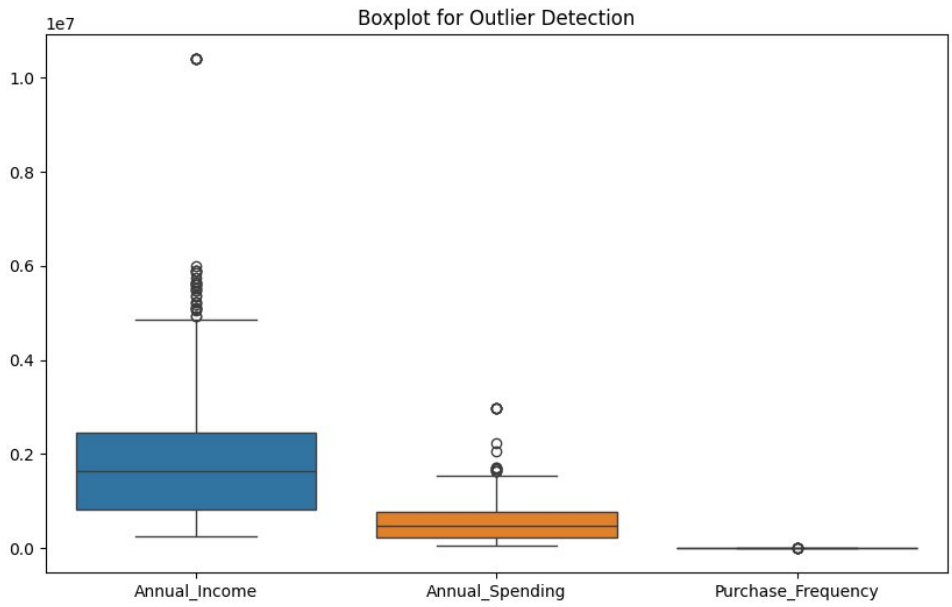


Fig.2 Graph from Phase 2 results

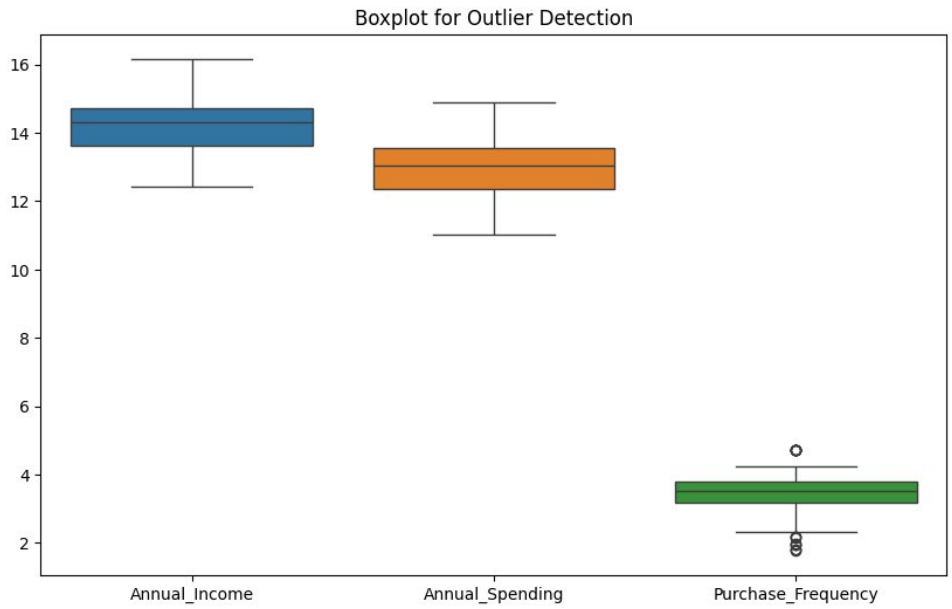


Fig.3 Graph from Phase 2 results

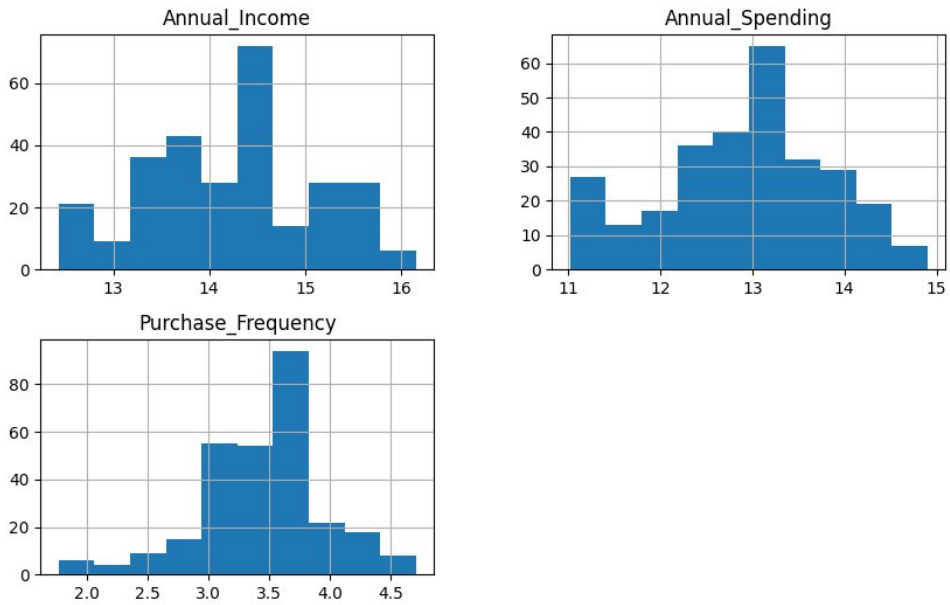


Fig.4 Graph from Phase 2 results

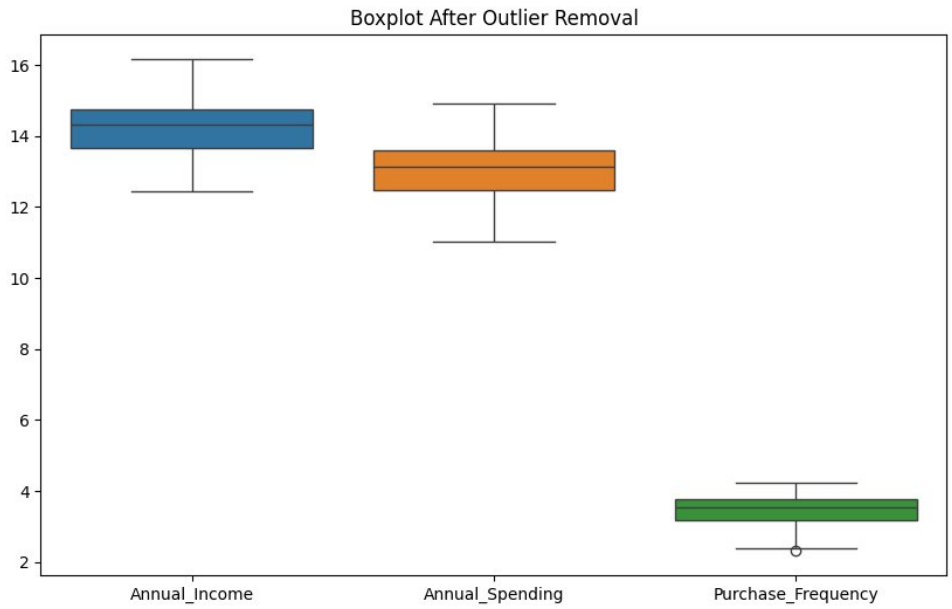


Fig.5 Graph from Phase 2 results

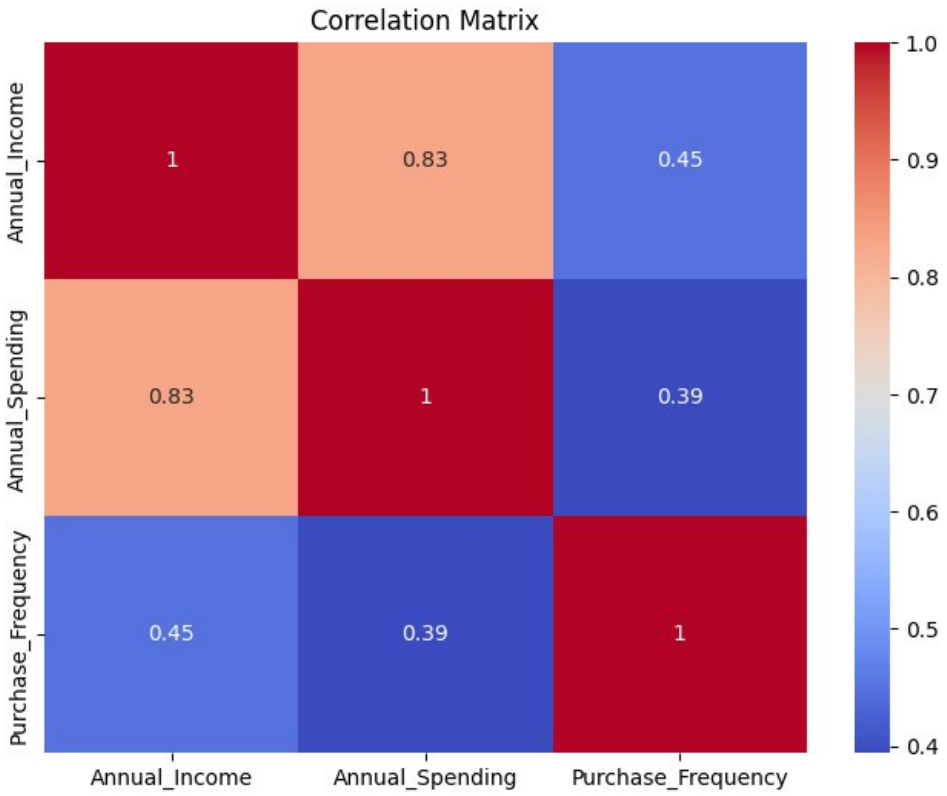


Fig.6 Heatmap showing corelation.

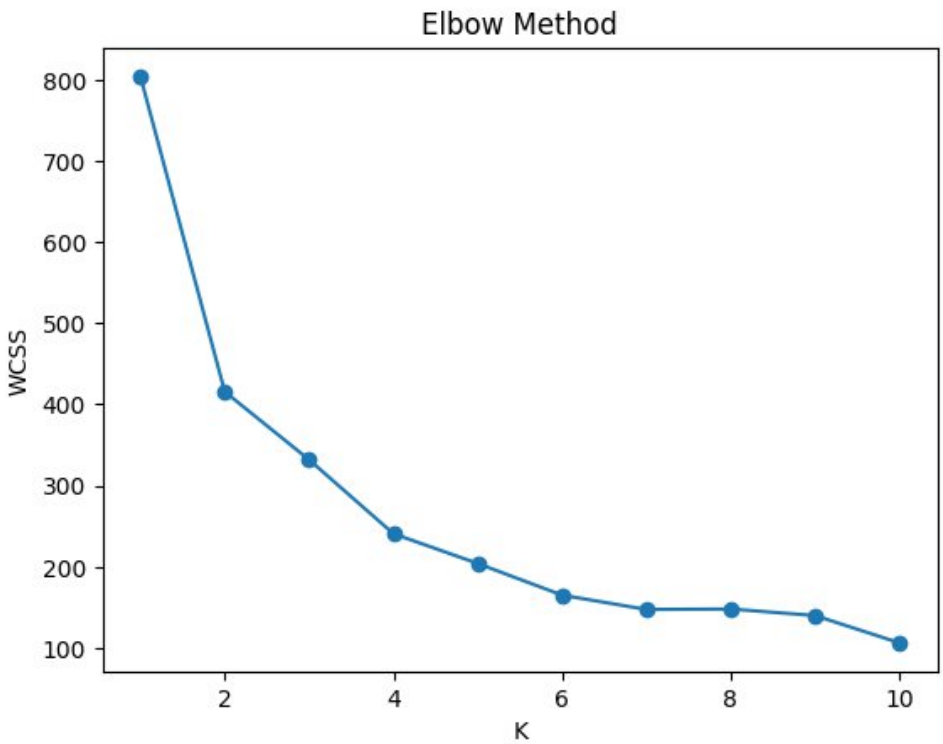


Fig.7 Graph from Phase 2 results

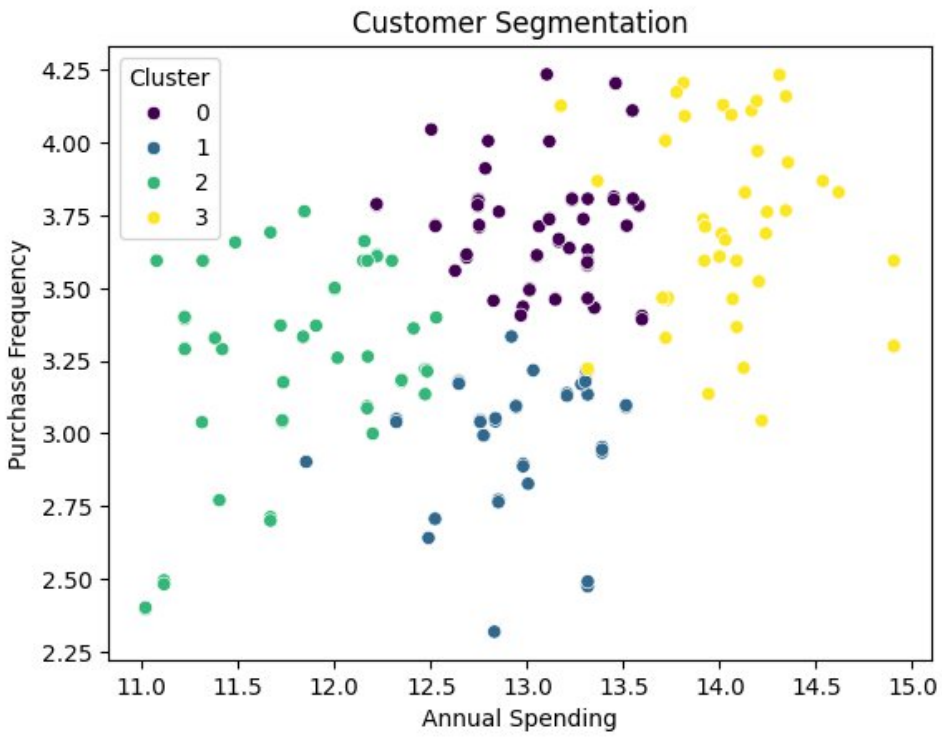


Fig.8 Graph from Phase 2 results

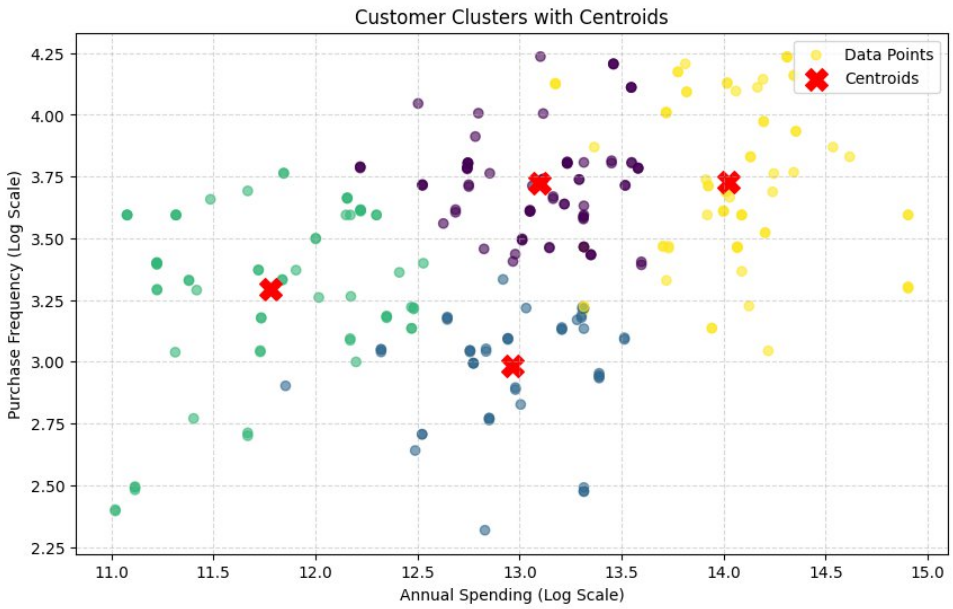


Fig.9 Graph from Phase 2 results

Fig.10 Graph from Phase 2 results

6. References:

Books:
1. Let Us Python – Yashavant Kanetkar
Online Resources:
• GeeksforGeeks – https://www.geeksforgeeks.org/
• TutorialsPoint – https://www.tutorialspoint.com/cprogramming/index.htm
• Stack Overflow – https://stackoverflow.com